* mandatory fields

You can either save or save and activate your changes.

3. (Optional): Provide the following information under **Custom Headers**:

Fields	Explanation	Usage (as maintained in Manage Credential Details)
Key Value	Additional information provided by you to your business partners or network partners as key value pairs. It will be sent as header parameters in a Consumer API. **Note** This feature is only available for SAP Business Network for Logistics standard APIs and not for custom API integrations.	Depending on your scenario, you can agree with your business partners or network partners to provide additional information.

4. Save your entries.

Manage Credential Details

Context

The **Manage Credential Details** view is a secure storage for you to maintain all the credentials that you receive from your business partner or network partner with whom you would like to enable B2B communication. A network partner can consist of a network of business partners (in this case of carriers) that provides services like tracking, freight tendering, and subcontracting.

SAP Business Network for Logistics supports the following credential types:

- Basic
- OAuth 2.0
- API Key
- project44

  For more information, see the Configuration and Application Guide for project44 add-ons.

Procedure

1. In the **Configure Partner Connections** app, choose **Manage Credential Details**.
2. Select the tab with the credential type that you want to use for maintaining the credential details.
3. Choose **Add** and complete the dialog.
4. (Optional): To change the credential details, select the row and choose **Edit**.

System Connections (Shipper)

Use the **System Connections** app for shippers to create a connection and to authenticate inbound and outbound messages between SAP Business Network for Logistics and your on-premise SAP Transportation Management standalone system or Transportation Management embedded in SAP S/4HANA.

ℹ Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

You can use the app in combination with each of the following exchange infrastructures between SAP Business Network for Logistics and your SAP backend system:

- SAP Cloud Integration
- SAP Process Integration
- Direct integration using SOA Manager

The process of using the System Connections app to create a connection and to authenticate messages includes the following steps:

1. Creating a Connection.
2. Authenticating Inbound Messages to SAP Business Network for Logistics.
3. Authenticating Outbound Messages from SAP Business Network for Logistics.
4. Activating the Connection.

Creating a Connection

By creating a connection in the System Connections app, you can then specify the details required for authenticating inbound and outbound messages between SAP Business Network for Logistics and your SAP backend system. If you need to connect multiple backend systems, create a connection for each system.

Procedure

1. In the **System Connections** app, click **Add** in the **Connections** column.
2. In the **New Connection** dialog box, select a connection type from the list.

   The following connection types are available:

   - **ABAP-based SAP system**
   - **Qualtrics**
   - **SAP Analytics Cloud**
   - **SAP Cloud ALM**
3. Enter the following information for the new connection.
   a. **Connection ID**: meaningful name of the connection
   b. **System ID**: ID of the system you're connecting, for example, QW9.

ℹ Note

Specify the source system ID as it appears in your external system (such as SAP Transportation Management). The format must be specified exactly as it appears in the external system. For example, QM7_910.

If you use another system ID or change the system ID afterwards, this can have severe consequences. Your processes might stop working.

⚠ Caution

If you use both the freight collaboration and global track and trace for tracking harmonization purposes, you must enter the system ID in the following format: *<System>_<Client ID>*. For example, QW9_170.

c. **Logical System ID**: logical system ID of the system you're connecting, for example, QW9CLNT170.

The **Logical System ID** field is only available to the ABAP-based SAP system connection type. This setting helps users using both the freight collaboration and global track and trace solutions avoid duplicated tracking records.

4. Click **Add**.

Related Information

Authenticating Inbound Messages to SAP Business Network for Logistics
Maintaining Qualtrics Connection
Maintaining SAC Connection
Maintaining SAP Cloud ALM Connection

Authenticating Inbound Messages to SAP Business Network for Logistics

To authenticate inbound messages to SAP Business Network for Logistics from your SAP backend system, complete the following steps in the System Connections app.

Prerequisites

You've acquired a client certificate that's signed by SAP or a third-party certification authority:

- You're recommended to use an SAP client certificate, which you can request from your Identity Authentication account on SAP BTP. The certificate is signed by SAP Passport Certification Authority. For more information, see Configuring a Certificate for API Authentication.
- Alternatively, you can use a client certificate that's signed by one of the following third-party certification authorities.

Third-party certification authorities

- AddTrust External CA Root
- Baltimore CyberTrust Root
- Certum CA
- DigiCert High Assurance EV Root CA
- DigiCert Global Root CA
- GeoTrust Global CA
- GeoTrust Primary Certification Authority - G3
- GlobalSign
- GlobalSign Root CA
- Go Daddy Class 2 Certification Authority
- Go Daddy Root Certificate Authority - G2
- Entrust Root Certification Authority

- Entrust.net Certification Authority (2048)
- Entrust Root Certification Authority - G2
- Equifax Secure Certificate Authority
- QuoVadis Root CA 2
- TC TrustCenter Class 2 CA II
- SwissSign Gold CA - G2
- SwissSign Platinum CA - G2
- SwissSign Silver CA - G2
- Thawte Primary Root CA
- Thawte Primary Root CA - G3
- VeriSign Class 1 Public Primary Certification Authority - G3
- VeriSign Class 3 Public Primary Certification Authority
- VeriSign Class 3 Public Primary Certification Authority - G5
- VeriSign Universal Root Certification Authority

Note

The System Connections app only supports CER format certificate files. If you have a certificate in a different format, you must convert it to a CER format file.

Context

When you have a client certificate, complete the following steps to authenticate inbound messages to SAP Business Network for Logistics.

Procedure

1. In the **System Connections** app, select the connection and on the **Inbound to Network** tab, click **Add**.
2. In the **Add Certificate** dialog, enter a meaningful name for the certificate details, and upload the client certificate.
    a. Note, you can use a certificate for only one connection.
    b. Make sure you upload the same client certificate to your SAP backend system. This step establishes trust between the systems.
3. When the certificate expires, you need to get a new one and repeat these steps to upload it.

    To receive notifications about certificate expiration, subscribe to the system notification service. For more information, see Subscribing to System Notifications.

Related Information

Authenticating Outbound Messages from SAP Business Network for Logistics

Authenticating Outbound Messages from SAP Business Network for Logistics

The process of authenticating outbound messages from SAP Business Network for Logistics to your on-premise SAP Transportation Management standalone system or Transportation Management embedded in SAP S/4HANA includes two steps:

- Mapping the URLs of your on-premise system to the service interfaces (message types) you want to use.
- Securing the communication between SAP Business Network for Logistics and your on-premise system with Client Certificate, Basic Authentication, or using Cloud Connector with Basic Authentication.

Related Information

Mapping the URLs of your SAP On-Premise System to Service Interfaces
Securing Communication Between SAP Business Network for Logistics and your SAP On-Premise System

Mapping the URLs of your SAP On-Premise System to Service Interfaces

The first step in authenticating outbound messages from SAP Business Network for Logistics to your on-premise SAP Transportation Management standalone system or Transportation Management embedded in SAP S/4HANA is to map the URLs of your on-premise system to the service interfaces (message types) you want to use.

Procedure

1. In the **System Connections** app, select the connection and then on the **Outbound from Network** tab, click **Add Destination**.
2. In the **Add Destination** dialog, select a service interface and enter the corresponding URL of your on-premise system (see the following table for more information).

Interface (Message)	Direction	Purpose
Default	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive all messages (listed in this table) from the SAP Business Network for Logistics tenant.
TransportationOrderQuotationConfirmation_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive freight quotations from the SAP Business Network for Logistics tenant.
TransportationOrderConfirmation_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive freight order or direct tendering confirmations from the SAP Business Network for Logistics tenant
InvoiceRequest_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive carrier invoices from the SAP Business Network for Logistics tenant
TransportationEventBulkNotification_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive event notifications for freight orders from the SAP Business Network for Logistics tenant
TransportationOrderChargeElementRequest_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive freight-order-charge updates after dispute resolution from the SAP Business Network for Logistics tenant
TransportationDisputeCaseNotification_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive dispute case notifications from the SAP Business Network for Logistics tenant

Interface (Message)	Direction	Purpose
GenericGateProcess_Out	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive gate process updates from the SAP Business Network for Logistics tenant
TransportationOrderBookingConfirmation_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive confirmations for freight bookings from the SAP Business Network for Logistics tenant.
TransportationOrderBookingConfirmation_In – Attachments	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive confirmations for freight bookings with attachments from the SAP Business Network for Logistics tenant.
BNAttachmentForConfirmation_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive attachments for a freight booking from the SAP Business Network for Logistics tenant.
TransportationOrderGenericRequest_In	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To create or update generic transportation order objects from SAP Business Network for Logistics tenant for external planning scenarios.
GenericAppointment_Out	Outbound from the SAP Business Network for Logistics tenant / Inbound to the on-premise system.	To receive dock appointment confirmation from the SAP Business Network for Logistics tenant.

For more information about supported releases with SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA, see SAP Note 2620322.

3. Optional: Select the authentication details. This is required if you're using Cloud Connector.
4. Optional: If you're using Cloud Connector, ensure that your URL starts with `http`.

> **Note**
>
> The Cloud Connector only supports inbound connections using HTTP; it doesn't support HTTPS. Administrators can choose between using HTTP or HTTPS when the Cloud Connector communicates with on-premise systems. Please refer to Supported Protocols of Cloud Connector for more details.
>
> If you select Cloud Connector with **Basic Authentication** as the **Active Authentication Type**, remember that you should configure the destination service URL with only the HTTP protocol. This URL corresponds to the virtual host set in your Cloud Connector's **Cloud To On-Premise** notification. This applies regardless of whether you've configured an HTTP or HTTPS port in your backend SAP S/4HANA system using SOAMANAGER configurations.

5. Repeat steps 2 to 4 for each service interface you want to use.

Related Information

Securing Communication Between SAP Business Network for Logistics and your SAP On-Premise System

The second step in authenticating outbound messages from SAP Business Network for Logistics to your on-premise SAP Transportation Management standalone system or Transportation Management embedded in SAP S/4HANA is to secure the connection with either client certificate authentication or Cloud Connector.

- Using client certificate authentication can be used with all exchange infrastructures, however, you must use it if you use the exchange infrastructure SAP Cloud Integration.

  Note, to use client certificate authentication, you must add a range of your system's IP addresses to an allowlist. For more information about adding IP addresses to an allowlist, see Country/Region Availability.

- Using basic authentication is possible in case your integration layer supports basic authentication.

- Using Cloud Connector is recommended. For more information about the advantages of using Cloud Connector, see Cloud Connector.

  Note, Cloud Connector is incompatible with the exchange infrastructure SAP Cloud Integration.

Related Information

Using Client Certificate Authentication
Using Cloud Connector with Basic Authentication
Using Basic Authentication

Using Client Certificate Authentication

To apply a client certificate, follow these steps:

Procedure

1. Open the **System Connections** app, locate the connection for which you want to apply or renew the client certificate, and then switch to the **Outbound from Network** tab on the right pane.
2. Choose **Configure Connections**.
    a. In the **General** tab, double confirm the **Active Authentication Type** is **Client Certificate**.
    b. Click **Download Certificate Chain** to download the client certificate (p7b files). If the expiration of the certificate is distant, the active certificate (CertChain_Active_TimeStamp.p7b) is downloaded. If the certificate is about to expire, both the active certificate and the new certificate (CertChain_New_TimeStamp.p7b) are downloaded. Both certificates are functional.
3. Extract the p7b file using KeyStore Explorer. We recommend extracting the new certificate if it's downloaded. Otherwise, extract the active one.
4. Locate the leaf certificate in the **Certificate Hierarchy** section and click **Export** to export it as a .cer file.
5. Open the .cer file in a text editor and copy the content.
6. Go to the SAP BTP cockpit, choose the subaccount that host your SAP Cloud Integration application.
7. Create an integration instance:
    a. Select **Process Integration Runtime** in the **Service** dropdown list, select **Integration-flow** in the **Plan** dropdown list and then specify the instance name.
    b. Set **Roles** to **ESBMessaging.send**, set **Grant-types** to **Client Credentials**, and then click **Create**.
8. In the newly created integration instance, click **Create** in the **Service Key** section to create a service key that applies the new client certificate:
    a. Set the **Key Type** to **External Certificate**.
    b. Paste the content you copied in step 5 into the **External Certificate** field.
    c. Save the service key.

Related Information

Securing Communication Between SAP Business Network for Logistics and your SAP On-Premise System
Activating the Connection

Using Cloud Connector with Basic Authentication

SAP Business Network for Logistics can be used in combination with Cloud Connector to enable secure communication with your SAP on-premise system (in this case SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA).

Prerequisites

- You have an SAP user account, for example an S-user or P-user in SAP ID service (https://accounts.sap.com).
    - The S-user ID is provided in the order form to your SAP Account Manager.
    - To obtain a P-user ID, go to https://www.sap.com/index.html and register a new user using his or her email address.
    - If a user has already been registered with the SAP ID service (for example, at SAP Community or SAP Developer Center) using the same email address, the user does not need to register again.

  For more information about creating a user, see Create SAP User Accounts.
- You have installed Cloud Connector.

  For more information about installing Cloud Connector, see Installation.

Context

After completing the prerequisites, complete the following steps in the System Connections app:

Procedure

1. Specify a location ID and SAP User to use with the Cloud Connector.
    a. Click **Manage Cloud Connector Details**.
    b. On the **Locations** tab, click **Add** and enter a location ID. You can also search for a location ID.

    Later on, you'll need to enter the location ID when adding the SAP Business Network for Logistics subaccount to Cloud Connector (see Next Steps below).

    c. Set the status to **Released**. This allows you to assign the location ID to an interface.
    d. On the **SAP Users** tab, click **Add**. In the **SAP User ID or Email ID** column, enter an email ID or the SAP user that you created earlier (see Prerequisites above). You can search also for an SAP user ID, or email ID.
    e. Click **Save**.
2. Configure your connection.
    a. Select the **Configure Connections** button in the **Outbound from Network** tab.
    b. In the **General** tab, select **Cloud Connector with Basic Authentication** from the **Active Authentication Type** drop down list.
    c. Select **Email ID** and **Location ID** from the corresponding dropdown lists.
    d. Click **Save**.

Next Steps

Add the SAP Business Network for Logistics subaccount to Cloud Connector with the following information:

Step 1: Creation of a subaccount

For customers in EU data center, the configuration is as below:

- **Region**: Europe (Frankfurt) - AWS
- **Subaccount**: 519bb3be-97e0-42bc-a0bd-28e11b695c12
- **Subaccount User**: Enter the SAP user that you provided in step 2c
- **Password**: Enter the password for the SAP user
- **Location ID**: Enter the location ID that you entered in step 1b

Step 1: Creation of a subaccount

For customers in US data center, the configuration is as below:

- **Region**: US West (WA)
- **Subaccount**: 9d94f93a-659d-4428-a5ac-d523773fb6db
- **Subaccount User**: Enter the SAP user that you provided in step 2c
- **Password**: Enter the password for the SAP user
- **Location ID**: Enter the location ID that you entered in step 1b

Step 2: Maintain Cloud to On-Premise configuration

1. Navigate to the **Cloud To On-Premise** UI within the newly created subaccount (as in Step 1).
2. On the **ACCESS CONTROL** tab under **Mapping Virtual to Internal System**, add a new system mapping with the following parameters:
    - **Back-end Type**: either an ABAP system or an SAP Process Integration system (depending on your system landscape)
    - **Protocol**: HTTP or HTTPS (based on your web service port definition)
    - **Internal Host** and **Internal Port**
    - **Virtual Host** and **Virtual Port** for ABAP system
    - **Principal Type**: None
3. Save your entries.
4. Under **Resources Accessible On**, add a resource for the new mapping with the following parameters:
5.
    - **URL Path**: / (or specific to your service URL path)
    - **Enabled**: True
    - **Access Policy**: Path and all sub-paths
6. Save your entries.
7. Under **Mapping Virtual To Internal System** in the **Action** column, click on the first icon on the left to check the availability. The **Check Result** column of the internal system should be **Reachable**.

ℹ Note

The Cloud Connector only supports inbound connections using HTTP; it doesn't support HTTPS. Administrators can choose between using HTTP or HTTPS when the Cloud Connector communicates with on-premise systems. Please refer to Supported Protocols of Cloud Connector for more details.

If you select Cloud Connector with **Basic Authentication** as the **Active Authentication Type**, remember that you should configure the destination service URL with only the HTTP protocol. This URL corresponds to the virtual host set in your Cloud Connector's **Cloud To On-Premise** notification. This applies regardless of whether you've configured an HTTP or HTTPS port in your backend SAP S/4HANA system using SOAMANAGER configurations.

For more information about adding a subaccount to Cloud Connector, see Managing Subaccounts.

Related Information

Securing Communication Between SAP Business Network for Logistics and your SAP On-Premise System
Maintaining Qualtrics Connection
Maintaining SAC Connection
Activating the Connection

Using Basic Authentication

SAP Business Network for Logistics can be used in combination with Cloud Connector to enable secure communication with your SAP on-premise system (in this case SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA). In case your integration layer supports Basic Authentication, you may use this option.

Prerequisites

- You have an integration layer between LBN and your on-premise system.
- You have done the required configuration in your integration layer to connect to LBN.

Procedure

1. In the **Outbound from Network** tab, choose **Configure Connections**.
2. In the **General** tab, choose **Active Authentication Type** as **Basic Authentication**.
3. Enter the basic credentials to your integration layer and save.

Maintaining Qualtrics Connection

You can maintain the system URL and the credentials for a connection to Qualtrics so that SAP Business Network for Logistics can trigger Qualtrics surveys.

Context

In the **System Connections** app, proceed as follows:

Procedure

1. To add a connection, choose **Add** in the **Connections** view. Select **Qualtrics** as the connection type, enter a connection ID and a system ID.
2. Choose **Maintain Authentication Details**, then **Add**. Provide an alias and an API key value for Qualtrics. The API key attribute is preset to X-API-TOKEN. Save your entries.
3. In the main view, maintain the system URL for Qualtrics and the authentication alias. Save your entries.
4. Activate the connection.